Interview Q&A 1

Forward Deployed Engineer – Kinetic IT

Technical Questions

Q1. Tell me about your technical background.

Answer:
I am currently studying software development at Charles Darwin University. Through my academic and practice projects, I have worked with JavaScript, TypeScript, Python, C#, SQL, React, Vue.js, Angular, ASP.NET Core, Entity Framework Core and SQL Server. I have also completed learning activities involving Node.js, REST APIs, Azure, AWS, Google Cloud, WebSockets, LLMs and RAG. My main full-stack project was a Task Manager where I connected multiple frontend frameworks to the same ASP.NET Core API and SQL Server database.

Q2. What is a REST API, and how have you used one?

Answer:
A REST API allows different applications to communicate through HTTP requests. In my Task Manager project, I created an ASP.NET Core REST API with GET, POST, PUT and DELETE endpoints. I tested the endpoints using Postman and then connected React, Vue.js and Angular frontends to the same API.

Q3. Explain the architecture of your Task Manager project.

Answer:
The architecture had three main layers. React, Vue.js or Angular provided the frontend interface, ASP.NET Core provided the REST API, and Entity Framework Core connected the API to SQL Server. This meant I could change the frontend framework without redesigning the backend or database.

Q4. What is Entity Framework Core?

Answer:
Entity Framework Core is an ORM for .NET. It allows C# objects and classes to be mapped to database tables and reduces the amount of SQL that needs to be written manually. In my project, I used a DbContext and DbSet<TaskItem> and methods such as FindAsync, ToListAsync and SaveChangesAsync.

Q5. What is the difference between frontend and backend development?

Answer:
The frontend is the part of the application that users interact with, while the backend handles application logic, APIs and communication with databases. In my practical project, React, Vue.js and Angular were frontend technologies, while ASP.NET Core and Entity Framework Core formed the backend and data-access layers.

Q6. What experience do you have with SQL?

Answer:
I completed SQL tutorials and practical exercises involving SELECT, SELECT TOP, DISTINCT, WHERE, ORDER BY, INSERT, UPDATE, DELETE, NULL values and AND, OR and NOT conditions. I then applied SQL Server in my C# Task Manager. Initially I accessed the database directly using Microsoft.Data.SqlClient, and later I used Entity Framework Core.

Q7. How would you troubleshoot an application that cannot communicate with an API?

Answer:
I would first determine which layer is failing. I would test the API directly using Postman or the browser to see whether the backend works independently. Then I would inspect the frontend request, URL, HTTP method and browser console. I would also check CORS configuration, network requests, server logs and database connectivity. I used this approach when I encountered CORS problems while connecting my frontend applications to ASP.NET Core.

Q8. What is CORS, and why did you need it?

Answer:
CORS stands for Cross-Origin Resource Sharing. Browsers restrict requests between applications running on different origins unless the server allows them. My Vue frontend ran on port 5173 and Angular on port 4200 while the API ran on port 5227, so I configured an ASP.NET Core CORS policy to allow those frontend origins.

Q9. What cloud platforms have you worked with?

Answer:
As part of my academic training, I explored Microsoft Azure, AWS and Google Cloud. My

experience is currently foundational rather than production-level, but the training helped me understand cloud services, deployment concepts and how cloud platforms fit into modern software systems.

Q10. What do you know about LLMs and RAG?

Answer:
I completed introductory learning around large language models and Retrieval-Augmented Generation. My understanding is that RAG combines a language model with retrieved external information so the model can use relevant information when generating its response. This was exploratory academic work rather than production AI development.

Q11. What are WebSockets?

Answer:
WebSockets provide persistent, two-way communication between a client and server. Unlike a typical request-response pattern where the client repeatedly sends HTTP requests, WebSockets are useful when applications need real-time communication, such as live notifications, chat or rapidly changing application data.

Q12. You don't know a technology required for a project. What would you do?

Answer:
I would first understand what the technology needs to accomplish and identify the core concepts I need. I would use documentation and tutorials, build a small practical example, test it independently, and then integrate it into the larger system. That is similar to how I approached unfamiliar technologies throughout my program, where I moved from tutorials into practical implementations.

Behavioural Questions – Forward Deployed Engineer

Q13. Tell me about a technical problem you encountered and how you solved it.

Answer:
While connecting my Vue frontend to my ASP.NET Core API, the browser blocked the requests because of CORS. I confirmed that the API itself worked, which helped me isolate the issue to frontend-to-backend communication. I then configured a CORS policy in ASP.NET Core to allow the frontend origin, restarted the API and tested the application again. Later, when I added Angular, I updated the configuration to allow both frontend development servers.

Q14. Tell me about a time you had to learn something quickly.

Answer:
During my role-based practical work, I had to move through several technologies including C#, ASP.NET Core, Entity Framework Core, React, Vue.js and Angular. I approached each technology by first learning its core concepts and then immediately applying them to a small practical. Eventually I reused the same Task Manager backend with three different frontend frameworks, which helped me understand both their differences and their common principles.

Q15. Tell me about a mistake or unexpected issue you encountered.

Answer:
When working with Angular, my application initially produced a blank page with an NG0908 error related to Zone.js. Instead of rebuilding the project, I checked the runtime error and configuration. I found that the generated Angular project was using a newer zoneless configuration, but my application configuration included a setting that expected Zone.js. I removed that conflicting configuration and the application worked correctly.

Q16. How do you approach a problem you haven't seen before?

Answer:
I try to break the problem into smaller parts rather than changing everything at once. For a full-stack application, for example, I would check the database, backend API and frontend independently. Once I know which layer is failing, I can focus my troubleshooting there. This approach helped me resolve several issues during my practical projects.

Q17. Why are you interested in a Forward Deployed Engineer role?

Answer:
I am interested in work that combines software development with solving practical problems. During my studies, I particularly enjoyed connecting different technologies, troubleshooting integration problems and learning unfamiliar tools when they were needed for a project. A Forward Deployed Engineer role interests me because it would allow me to continue developing those capabilities while working on real systems and user requirements.

Q18. What is one area you still need to develop?

Answer:
Most of my current experience comes from academic and practice projects rather than large production systems. I have built working full-stack applications and explored cloud technologies, but I still need more experience with production deployment, larger codebases and team-based development. Those are areas I want to develop as I move into professional software engineering.